

Probability Distributions

2.1 INTRODUCTION

Under given conditions, observed (or empirical) frequency distributions can be approximated by theoretical distributions. The theoretical distributions: (a) provide a logical basis for making decisions (b) are useful in making predictions on the basis of limited informations or theoretical considerations (c) indicate as what to expect if the *random variable* behaves as we assume it does. The probability distributions are *discrete or continuous* according as the random variable assumes discrete or continuous values. The most important discrete distribution are *binomial* and *poisson* distribution. *Normal distribution* is one of the most important continuous distribution. In this chapter we will discuss these three main probability distributions.

2.2 BINOMIAL DISTRIBUTION

Binomial distribution is one of the widely used probability distribution of a discrete random variable. The binomial distribution describes discrete data resulting from an experiment called a *Bernoulli process*. For example, the tossing of a fair coin a fixed number of times is a Bernoulli process, and the outcomes of such tosses can be represented by a *binomial probability distribution*. We can describe Bernoulli process as follows:

1. Each trial has only two possible outcomes : e.g. yes or no, success or failure, heads or tails
2. The probability of a success on any trial remains fixed over time. In case of tossing of a fair coin (trial), the probability of heads remains 0.5 for every toss no matter how many times the coin is tossed.
3. The trials are statistically independent i.e. in tossing of a fair coin, the outcome of one toss has no effect on the outcome of any other toss.

2.2.1 Definition of Binomial Distribution

If a trial of an experiment can result in success with probability p and failure with probability q ($= 1 - p$), then the probability of exactly x successes in n independent trials is given by

$$P(X = x) = f(x) = {}^nC_x p^x q^{n-x} \quad (2-1)$$

where $x = 0, 1, 2, \dots, n$; ${}^nC_x = \frac{n!}{x!(n-x)!}$; and $p + q = 1$

The equation (3-1) is called the binomial probability distribution (or function) with *parameters* p and n .

The entire probability distribution of $0, 1, 2, \dots, n$ successes are respectively $q^n, {}^nC_1 q^{n-1} p, {}^nC_2 q^{n-2} p^2, \dots, p^n$, which are the successive terms in the binomial expansion $(q + p)^n$.

2.2.2. Properties of Binomial Distribution

1. The binomial distribution is completely known if n and p are known.
2. Since the random variable X takes integral values $0, 1, 2, \dots, n$, the binomial distribution is a discrete probability distribution.
3. If $p = q = \frac{1}{2}$, the binomial distribution is symmetrical.

Even if $p \neq q$, the binomial distribution tends to symmetrical distribution when n tends to sufficiently large value.

If $p < 1/2$, it is skewed to the right (or positively skewed).

If $p > 1/2$, it is skewed to the left (or negatively skewed).

4. If np is a whole number, the mean and mode of the binomial distribution are equal, and the distribution becomes *unimodal*.
5. The mean and the standard deviation of the binomial distribution are

$$\text{Mean; } \mu = np \quad (2-2)$$

$$\text{S.D.; } \sigma = \sqrt{npq} \quad (2-3)$$

2.2.3 Conditions for Using Binomial Distribution

The binomial distribution can be used under the following conditions.

- The random experiment should be performed repeatedly finite number of times i.e. *n should be finite*.
- The result of any trial can be classified under two mutually exclusive categories called *success and failure*.
- The probability of success in each trial should remain constant
i.e. *p should be constant*
- The trials should be *independent* so that the result of any trial is unaffected by the result of previous trials.

2.2.4 Fitting of Binomial Distribution

If a random experiment consisting of *n* trials is repeated *N* times satisfying the conditions of the binomial distribution then the frequency of *x* successes is given by

$$g(x) = N \cdot f(x) = N \cdot {}^nC_x p^x q^{n-x} \quad (2-4)$$

Putting *x* = 0, 1, 2,, *n* successively we get the expected or theoretical frequencies of the binomial distribution as follows :

$$N q^n, N \cdot {}^nC_1 q^{n-1} p, N \cdot {}^nC_2 q^{n-2} p^2, \dots, N p^n$$

If *p* is unknown, it can be estimated by using the relation

$$np = \frac{\sum fX}{N}$$

Then we can fit a binomial distribution to a given frequency distribution by using the formula (2-4).

Example 1.

A manufacturing process turns out articles that on the average 10% defective. Compute the probability of 0, 1, 2, and 3 defective articles that might occur in the sample of 3 articles.

Solution :

Prob. of a defective article is : $p = 10\% = 0.1$; $q = 1 - p = 1 - 0.1 = 0.9$. The probability of *x* defectives in a sample of 3 articles is given by the binomial probability law

$$f(x) = {}^3C_x p^x q^{3-x} = {}^3C_x (0.1)^x (0.9)^{3-x}$$

where $x = 0, 1, 2, 3$

Put $x = 0, 1, 2, 3$ successively, we get the probability of 0, 1, 2, 3 defective articles as :

$$f(0) = (0.9)^3 = 0.729$$

$$f(1) = {}^3C_1 (0.1) (0.9)^2 = 3 \times 0.1 \times 0.81 = 0.243$$

$$f(2) = {}^3C_2 (0.1)^2 (0.9) = 3 \times 0.01 \times 0.9 = 0.027$$

$$f(3) = (0.1)^3 = 0.001$$

Example 2.

Three perfect coins are tossed together. What is the probability of getting atleast one head ?

Solution :

Let *x* be the number of heads obtained in a throw of three perfect coins. Then $p(x) = {}^3C_x p^x q^{3-x}$

where $p = \frac{1}{2}$ = probability of head in a throw of a coin,

$$q = 1 - p = \frac{1}{2}$$

Probability of getting atleast one head

= 1 - Probability of getting no head

$$= 1 - f(0) = 1 - {}^3C_0 (1/2)^0 (1/2)^{3-0} = 1 - \frac{1}{8} = \frac{7}{8}$$

Aliter: Reqd. prob. $P(x \geq 1) = f(1) + f(2) + f(3)$

$$= \frac{1}{8} ({}^3C_1 + {}^3C_2 + {}^3C_3) = \frac{7}{8}$$

Example 3.

An accountant is to audit 24 accounts of a firm. Sixteen of these are of highly - valued customers. If the accountant selects 4 of the accounts at random, what is the probability that he chooses at least one highly-valued account.

Solution :

Probability that the account is of highly-valued customer is :

$$p = \frac{16}{24} = \frac{2}{3} \quad \therefore q = 1 - p = 1 - \frac{2}{3} = \frac{1}{3}$$

The probability that in a random sample of size $n = 4$, the accountant chooses the accounts of x highly-valued customers is given by

$$f(x) = {}^4C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{4-x}$$

Probability that the accountant chooses at least one highly-valued account

$$= 1 - P(\text{no highly-valued account}) = 1 - f(0)$$

$$= 1 - {}^4C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{4-0} = 1 - \frac{1}{81} = \frac{80}{81}$$

Example 4.

The average percentage of failures in a certain examination is 40. What is the probability that out of a group of 6 candidates, at least 4 passed in the examination ?

Solution :

$$\text{Here } n = 6, q = 40\% = 0.4, p = 1 - q = 1 - 0.4 = 0.6$$

The probability that x candidates passed the examination is given by

$$f(x) = {}^6C_x (0.6)^x (0.4)^{6-x}$$

$$\text{Required Probability} = P(x \geq 4) = f(4) + f(5) + f(6)$$

$$= {}^6C_4 (0.6)^4 (0.4)^2 + {}^6C_5 (0.6)^5 (0.4) + {}^6C_6 (0.6)^6 \\ = 0.311040 + 0.186624 + 0.046656 = 0.544320$$

Example 5.

The incidence of occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers, 4 or more will contract the disease ?

Solution :

$$\text{Here, } n = 6, p = 20\% = 0.2 = \frac{1}{5}, \quad q = 1 - p = 1 - \frac{1}{5} = \frac{4}{5}$$

By the binomial probability law, the prob. that out of 6 workers, x will contract the disease is:

$$f(x) = {}^6C_x p^x q^{6-x} = {}^6C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{6-x} = \frac{1}{5^6} \cdot {}^6C_x \cdot 4^{6-x}$$

Hence the probability that in a sample of 6 workers 4 or more will contract the disease is:

$$f(4) + f(5) + f(6) = \frac{1}{5^6} ({}^6C_4 \times 4^2 + {}^6C_5 \times 4 + {}^6C_6) \\ = \frac{1}{5^6} (15 \times 16 + 6 \times 4 + 1) = \frac{265}{5^6} \\ = 0.01696$$

Example 6.

An insurance salesmen sells policies to 5 men, all of identical age and good health. According to the actuarial tables the probability that a man of this particular age will be alive 30 years hence is $2/3$. Find the probability that 30 years hence (i) at least one man will be alive. (ii) at least 3 men will be alive.

Solution:

$$\text{Here } n = 5, p = 2/3, q = 1 - 2/3 = 1/3.$$

By the binomial probability law

$$f(x) = P(x \text{ out of } n \text{ men will alive 30 yrs. hence}) \\ = {}^nC_x p^x q^{n-x} = {}^5C_x (2/3)^x (1/3)^{5-x}$$

$$(i) \text{ Req'd. prob.} = P(\text{at least one man will alive 30 yrs. hence})$$

$$= 1 - P(\text{no man will be alive}) = 1 - f(0)$$

$$= 1 - {}^5C_0 (2/3)^0 (1/3)^{5-0} = 1 - \left(\frac{1}{3}\right)^5$$

$$= 1 - \frac{1}{243} = \frac{242}{243}$$

$$(ii) \text{ Req'd. prob.} = P(\text{at least 3 men will alive}) = f(3) + f(4) + f(5)$$

$$= {}^5C_3 (2/3)^3 (1/3)^2 + {}^5C_4 (2/3)^4 (1/3) + {}^5C_5 (2/3)^5$$

$$= \frac{1}{3^5} ({}^5C_3 \times 2^3 + {}^5C_4 \times 2^4 + {}^5C_5 \times 2^5)$$

$$= \frac{192}{243} = \frac{64}{81}$$

Example 7.

Out of 1000 families of 3 children each, how many families would you expect to have two boys and one girl assuming that boys and girls are equally likely ?

Solution :

Here $N = 1000$, $n = 3$, $p = P(\text{boy}) = 1/2$, $q = P(\text{girl}) = 1/2$

The expected number of families who have x boys and $n - x$ girls is :

$$g(x) = N \cdot f(x) = N \cdot {}^nC_x p^x q^{n-x} = 1000 \cdot {}^3C_x (1/2)^x (1/2)^{3-x} \\ = 1000 \cdot {}^3C_x \left(\frac{1}{2}\right)^3$$

Here the expected number of families who have 2 boys and 1 girl is :

$$g(2) = 1000 \cdot ({}^3C_2) (1/2^3) = 1000 \times 3 \times \frac{1}{8} = 375$$

Example 8.

The mean of a binomial distribution is 4 and its standard deviation is $\sqrt{3}$. What are the values of n , p and q with usual notations ?

Solution :

We have, Mean = $np = 4$ (i)

S.D. = $\sqrt{npq} = \sqrt{3}$ (ii)

$\therefore$ Variance = $npq = 3$ (iii)

Divide (iii) by (i) : $q = \frac{3}{4}$ $\therefore p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$

Substitute the value for p in (i) : $n\left(\frac{1}{4}\right) = 4$ $\therefore n = 4 \times 4 = 16$

Example 9.

Five fair coins were tossed 100 times. From the following outcomes calculate the expected frequencies

No of heads	:	0	1	2	3	4	5
Observed frequency	:	2	10	24	38	18	8

Solution :

Here $n = 5$, $N = \Sigma f = 100$, $p = P(H) = 1/2$

$q = P(T) = 1 - p = 1 - 1/2 = 1/2$

By the binomial law the frequency of x heads in 100 tosses of 5 coins each is

$$g(x) = N \cdot f(x) = N \cdot {}^nC_x p^x q^{n-x} \\ = 100 \times {}^5C_x (1/2)^x (1/2)^{5-x} \\ = 100 \times {}^5C_x \left(\frac{1}{2}\right)^{x+5-x} \\ = \frac{100}{32} \cdot {}^5C_x = 3.125 \times {}^5C_x$$

Put $x = 0, 1, 2, \dots, 5$ successively we get the expected binomial frequencies as follows :

No. of heads	Expected	Binomial	Frequency
0	$3.125 \times {}^5C_0 =$	$3.125 \times 1 =$	$3.125 \approx 3$
1	$3.125 \times {}^5C_1 =$	$3.125 \times 5 =$	$15.625 \approx 16$
2	$3.125 \times {}^5C_2 =$	$3.125 \times 10 =$	$31.25 \approx 31$
3	$3.125 \times {}^5C_3 =$	$=$	$31.25 \approx 31$
4	$3.125 \times {}^5C_4 =$	$3.125 \times 10 =$	$15.625 \approx 16$
5	$3.125 \times {}^5C_5 =$	$=$	$3.125 \approx 3$
		$3.125 \times 5 =$	
		$3.125 \times 1 =$	

The expected binomial frequencies are :

3, 16, 31, 31, 16, 3

Example 10.

Fit a binomial distribution to the following data:

X :	0	1	2	3	4
f :	28	62	46	10	4

Solution :

Here $n = 4$, $N = \Sigma f = 150$

$$\text{Mean: } np = \frac{\Sigma fX}{N} = \frac{28 \times 0 + 62 \times 1 + 46 \times 2 + 10 \times 3 + 4 \times 4}{150}$$

$$= \frac{4}{3}$$

$$\text{or, } 4p = \frac{4}{3} \quad \therefore p = \frac{1}{3}$$

$$\therefore q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

By the binomial probability law, the expected frequencies are given by

$$g(x) = N \cdot f(x) = N \cdot {}^nC_x p^x q^{n-x}$$

$$= 150 \cdot {}^4C_x (1/3)^x (2/3)^{4-x}$$

$$= 150 \cdot {}^4C_x \left(\frac{1}{3}\right)^x \cdot \left(\frac{1}{3}\right)^{4-x} (2)^{4-x}$$

$$= 150 \cdot {}^4C_x \left(\frac{1}{3}\right)^4 (2)^{4-x}$$

$$= \frac{150}{81} \cdot {}^4C_x 2^{4-x} = \frac{50}{27} \cdot {}^4C_x (2^4 - x)$$

X	Expected frequencies
0	$\frac{50}{27} \times {}^4C_0(2^4) = \frac{50}{27} \times 1 \times 16 = 29.63 \approx 30$
1	$\frac{50}{27} \times {}^4C_1(2^3) = \frac{50}{27} \times 4 \times 8 = 59.26 \approx 59$
2	$\frac{50}{27} \times {}^4C_2(2^2) = \frac{50}{27} \times 6 \times 4 = 44.44 \approx 44$
3	$\frac{50}{27} \times {}^4C_3(2^1) = \frac{50}{27} \times 4 \times 2 = 14.82 \approx 15$
4	$\frac{50}{27} \times {}^4C_4(2^0) = \frac{50}{27} \times 1 \times 1 = 1.85 \approx 2$

Example 11.

A sampling scheme involves taking a sample of ten items from each batch produced and rejecting the batch if more than two defectives are found. If in fact five per cent of all items are defective, what is the probability that a batch will be rejected ?

Solution :

Here $p = 0.05$ probability that the item is defective.

$$n = 10$$

$$\therefore q = 1 - p = 1 - 0.05 = 0.95$$

We have to compute, using binomial distribution,

$$P(\text{batch rejected}) = P(3) + P(4) + P(5) + P(6) + P(7) + P(8) + P(9) + P(10)$$

$$P(x > 2) = 1 - P(2 \text{ or less defective})$$

$$= 1 - [P(0) + P(1) + P(2)]$$

$$\text{where } P(0) = {}^{10}C_0 p^0 q^{10-0} = q^{10} = (0.95)^{10} = 0.5987$$

$$P(1) = {}^{10}C_1 p q^{10-1} = 10 \times (0.05) \times (0.95)^9 = 0.3151$$

$$P(2) = {}^{10}C_2 p^2 q^{10-2} = \frac{10 \times 9}{1 \times 2} (0.05)^2 \times (0.95)^8 = 0.0746$$

$$\therefore P(\text{batch rejected}) = 1 - (0.5987 + 0.3151 + 0.0746)$$

$$= 1 - 0.9884 = 0.0116$$

Thus 1.16% of batches can be expected to be rejected by this sampling scheme.

EXERCISE 2 (A)

- What do you understand by Binomial distribution ? What are its features ?
- A lot containing 50 items has 5 items as defective. Indicate the probability of the following if 4 items are drawn with replacement :
 - exactly 2 defective items

- (ii) atleast 2 defective items
(iii) at the most 2 defective items
3. A coin is tossed six items. What is the probability of obtaining four or more heads.
4. Four coins are tossed simultaneously. What is the probability of getting
(i) 2 heads and 2 tails (ii) at least two heads, and (iii) at least one head ?
5. The incidence of a certain disease is such that on the average 20% of workers suffer from it. If 10 workers are selected at random, find the probability that
(i) exactly 2 workers suffer from the disease.
(ii) not more than 2 workers suffer from the disease.
Calculate the probability up to fourth decimal places.
6. The probability of a bomb hitting a target is $1/5$. Two bombs are enough to destroy a bridge. If six bombs are aimed at the bridge, find the probability that the bridge is destroyed.
7. Out of 320 families with 5 children each, how many families would be expected to have (i) 2 boys and 3 girls (ii) at least one boy ? Assume equal probability for boys and girls.
8. If hens of a certain breed lay eggs on 5 days a week on an average, find how many days during a season of 100 days a poultry keeper with 5 hens of this breed, will expect to receive at least 4 eggs ?
9. The screws produced by a certain machine were checked by examining samples of 12. The following table shows the distribution of 128 samples according to the number of defective items they contained ;
- | | | | | | | | | | |
|---|---|---|----|----|----|----|---|---|-------|
| No. of defectives
in a samples of 12 : | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total |
| No. of samples : | 7 | 6 | 19 | 35 | 30 | 23 | 7 | 1 | 128 |
- Fit a binomial distribution and find the expected frequencies if the chance of machine being defective is $1/2$. Find the mean and variance of the fitted distribution.

10. (a) A student obtained the following answer to a certain problem given to him. Mean = 2.4, variance = 3.2 for a binomial distribution. Comment on the result.
(b) Is there any inconsistency in the statement, the mean of binomial distribution is 40 and its standard deviation 4 ? If no, inconsistency is found. What shall be the values of p , q and n ?
11. Razor blades of a certain kind are sold in packets of five. The following table shows the distribution of 100 packets according to the number of faulty blades they contained.
- | | | | | | | |
|-------------------------|----|----|---|---|---|---|
| Number of faulty blades | 0 | 1 | 2 | 3 | 4 | 5 |
| Number of packets | 80 | 17 | 2 | 1 | 0 | 0 |
- Calculate the mean number of faulty blades per packet and, assuming a binomial distribution, the proportion of all blades produced which are faulty.
- Use your result to calculate the probabilities of 0, 1, 2, 3, 4 or 5 faulty blades in a packet and hence find the numbers of packets out of the total of 100 which could be expected to contain these respective numbers of faulty blades.
12. Fit a binomial distribution to the following data
- | | | | | | | |
|-------|---|----|----|----|----|---|
| $x :$ | 0 | 1 | 2 | 3 | 4 | 5 |
| $f :$ | 1 | 24 | 10 | 18 | 35 | 8 |
- Also find out the mean and variance of the distribution ?
13. One hundred and ninety two families (for each of which the probability of an albino being born is otherwise established) had the following distribution of albinos among the first three children.
- | | | | | |
|-----------------|----|----|----|---|
| No. of children | 0 | 1 | 2 | 3 |
| No. of families | 77 | 90 | 20 | 5 |
- Find the expected frequencies on the basis of a theoretical probability 0.25 of a child being born an albino.
14. Five fair coin were tossed 200 times. Find the expected frequencies of the following distribution.
- | | | | | | | |
|--------------|----|----|----|----|----|----|
| No. of heads | 0 | 1 | 2 | 3 | 4 | 5 |
| frequency | 12 | 28 | 60 | 56 | 30 | 14 |

15. Fit a binomial distribution and find the expected frequencies to the following data :

X :	0	1	2	3	4
f :	20	62	46	10	4

2.3 POISSON DISTRIBUTION

In binomial distribution it is possible to count the number of times an event is observed i.e. n is precisely known. There are some rare and causal events like accidents on a road, defects in a product, goals scored at a football match, the arrivals of patients at a health clinic, the number of looms waiting for service in a textile mill, the number of telephone calls going through a central switching system etc. in which we know the number of times an event occurs but not how many times it does not occur. It is clear that the total number of trials with regard to a given experiment are not precisely known. For such rare events, *Poisson distribution* is used. Note that all the above examples can be described by a discrete random variable that takes on non negative whole number values 0, 1, 2, 3, 4, 5 and so on.

2.3.1 Definition of Poisson Distribution

Poisson distribution is the limiting case of the binomial distribution under the conditions:

- $n \rightarrow \infty$ i.e. the number of trials is indefinitely large.
- $p \rightarrow 0$ i.e. the constant probability of success for each trial is indefinitely small.
- $np = \lambda$ i.e. the product np is a finite number say, λ .

The probability distribution (or function) of a random variable X which follows Poisson distribution is defined by

$$P(X = x) = p(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad (2-5)$$

where $x = 0, 1, 2, \dots$, $e = 2.71828$ (exponential constant)

The probability that the random variable X can take on non negative whole number values 0, 1, 2, 3, are respectively

$$e^{-\lambda}, \lambda e^{-\lambda}, \frac{\lambda^2 e^{-\lambda}}{2!}, \frac{\lambda^3 e^{-\lambda}}{3!}, \dots$$

$$\text{so that } e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2 e^{-\lambda}}{2!} + \frac{\lambda^3 e^{-\lambda}}{3!} + \dots \text{ to } \infty = 1$$

2.3.2 Characteristics of Poisson Distribution

- If the value of m is known, all the probabilities of the Poisson distribution can be determined. Thus m is called the *parameter* of the Poisson distribution.
- The Poisson distribution is based on the following assumptions :
 - the occurrence or non - occurrence of an event does not influence the occurrence or non-occurrence of any other event.
 - the probability of success for a short time interval (or a small space) is proportional to the length of the time interval (or space).
 - the probability of occurrence of more than one event in a very small interval is negligible.
- The mean and the standard deviation a Poisson distribution is given by

$$\text{Mean} = \lambda \quad (2-6)$$

$$S.D = \sqrt{\lambda} \quad (2-7)$$

2.3.3 Fitting of Poisson Distribution

The steps for fitting of Poisson distribution are as follows :

Step 1. Compute the mean $\bar{X}$ of the given distribution and take $\lambda = \bar{X}$

Step 2. Obtain the value of $e^{-\lambda}$ by using the calculator or the relation

$$e^{-\lambda} = \frac{1}{\text{antilog}(0.4343 \times \lambda)}$$

Step 3. Calculate the various probabilities of the Poisson distribution by using the formula :

$$P(X = x) = p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{where } x = 0, 1, 2, \dots, n$$

Step 4. List out the expected or theoretical frequencies according to Poisson distribution :

$$f(x) = N \cdot p(x) \quad \text{where } N = \text{total frequency} \quad (2-8)$$

Example 12.

Accidents occur on a particular stretch of highway at an average rate of 3 per week. What is the probability that there will be exactly two accident in a given week ?

Solution :

Let X be the number of accidents occurring on the highway. Then X has Poisson distribution with parameter $\lambda = 3$, the average number of accidents per week.

Hence the probability of r accidents per week is given by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!},$$

The probability of exactly two accidents in a given week is

$$\begin{aligned} P(X = 2) &= \frac{e^{-3} \times 3^2}{2!} = \frac{9}{2 \times e^3} = \frac{9}{2 \times (2.718)^3} \\ &= \frac{9}{2 \times 20.08} = 0.2241 \end{aligned}$$

Example 13.

Between the hours of 2 and 4 P.M. the average number of phone calls per minute coming into the switch-board of a company is 2.5. Find the probability that during one particular minute there will be (i) no phone call at all (ii) exactly 3 calls (iii) at least 5 calls. (Given $e^{-2} = 0.13534$ and $e^{-0.5} = 0.606540$)

Solution :

Let X be the number of telephone calls per minute. Then X will follow Poisson distribution with parameter $\lambda = 2.5$. The probability of x telephone calls per minute is given by

$$P(X = x) = \frac{e^{-2.5} \times (2.5)^x}{x!}; \quad x = 0, 1, 2, \dots$$

(i) The prob. of no phone call at all is :

$$P(X = 0) = \frac{e^{-2.5} \times (2.5)^0}{0!} = \frac{e^{-2.5} \times e^{0.5}}{1}$$

$$= 0.13534 \times 0.60650$$

$$= 0.082$$

$$\begin{aligned} \text{(ii) Prob. of exactly 3 calls is : } P(X = 3) &= \frac{e^{-2.5} (2.5)^3}{3!} \\ &= \frac{0.0821 \times 15.625}{6} = 0.2138 \end{aligned}$$

(iii) Probability of at least 5 calls is :

$$\begin{aligned} P(X \geq 5) &= 1 - P(X \leq 4) = 1 - [p(0) + p(1) + p(2) + p(3) + p(4)] \\ &= 1 - \left[e^{-2.5} + \frac{e^{-2.5} \times 2.5}{1!} + \frac{e^{-2.5} (2.5)^2}{2!} + \frac{e^{-2.5} (2.5)^3}{3!} + \frac{e^{-2.5} (2.5)^4}{4!} \right] \\ &= 1 - e^{-2.5} \left[1 + 2.5 + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!} + \frac{(2.5)^4}{4!} \right] \\ &= 1 - 0.0821 \left[1 + 2.5 + \frac{6.25}{2} + \frac{15.625}{6} + \frac{39.0625}{24} \right] \\ &= 1 - 0.0821 (1 + 2.5 + 3.125 + 2.6042 + 1.6276) \\ &= 1 - 0.0821 \times 10.8568 = 1 - 0.8913 \\ &= 0.1087 \end{aligned}$$

Example 14.

An automatic machine makes paper clips from coils of wire. On the average 1 in 400 paper clips is defective. If the paper clips are packed in boxes of 100, what is the probability that any given box of clips will contain (i) no defective (ii) one or more defectives (iii) less than two defectives. Given $e^{-0.25} = 0.7787$

Solution :

Here, p = prob. of a defective paper clip = $1/400$, $n = 100$

$$\therefore \lambda = n \times p = 100 \times \frac{1}{400} = 0.25$$

= mean number of defectives in a box

By using Poisson distribution we have the prob. of x defective clips in a box is :

$$P(X = x) = p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.25} \times (0.25)^x}{x!}$$

- (i) Req'd. prob. = $p(0) = \frac{e^{-0.25} (0.25)^0}{0!} = e^{-0.25} = 0.7787$
- (ii) Req'd. prob. = $p(X \geq 1) = 1 - p(0) = 0.2213$
- (iii) Req'd. prob. = $p(X < 2) = p(0) + p(1)$
 $= e^{-0.25} + e^{-0.25} \times 0.25 = e^{-0.25} (1 + 0.25)$
 $= 0.7787 (1 + 0.25)$
 $= 0.9734$

Example 15.

Find the probability that at most 5 defective bolts will be found in a box of 200 bolts if it is known that 2 per cent of such bolts are expected to be defective (you may take the distribution to be Poisson). Take $e^{-4} = 0.0183$

Solution :

Here $n = 200$, $p = \text{prob of defective bolt} = 2\% = 0.02$

$$\therefore \lambda = n \times p = 200 \times 0.02 = 4$$

= mean no. of defective bolts in the box

Using Poisson distribution the prob of x defective bolts in the box is :

$$P(X = x) = p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-4} (4)^x}{x!}$$

Hence the prob. that the box will contain at most 5 defective bolts is :

$$\begin{aligned} P(X \leq 5) &= p(0) + p(1) + p(2) + p(3) + p(4) + p(5) \\ &= e^{-4} \left[1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!} \right] \\ &= e^{-4} \left(1 + 4 + 8 + \frac{32}{3} + \frac{32}{3} + \frac{128}{15} \right) \\ &= 0.0183 (1 + 4 + 8 + 10.6667 + 10.6667 + 8.5333) \\ &= 0.7845 \end{aligned}$$

Example 16.

The management of photograph record company has discovered that the number of defects of records appears to follow a Poisson distribution with a mean equal to 0.4 (i) What is the probability that a record selected at random will have three defects? (ii) If management sets a policy that all photograph records sold to customers must not have any defects, what per cent of its records production will not be made available for sales because of defects ?

Solution :

Let X be a number of defects in a record. Then the prob. of r defects, by using Poisson prob. distribution with mean $X = 0.4$, is given by

$$P(X = x) = p(x) = \frac{e^{-0.4} (0.4)^x}{x!}$$

$$(i) \text{ Req'd. prob.} = p(3) = \frac{e^{-0.4} (0.4)^3}{3!} = \frac{0.6703 \times 0.064}{6} = 0.00715$$

- (ii) The photographic records will not be made available for sales to the customers even if it contains at least one defect and the required probability is given by

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - p(0) = 1 - e^{-0.4} \\ &= 1 - 0.6703 = 0.3297 \end{aligned}$$

Hence, about 33% of its photographic record production will not be made available for sales because of defects.

Example 17.

What probability model is appropriate to describe a situation where 100 misprints are distributed randomly throughout 100 pages of a book ? For this model, what is the probability that a page observed at random will contain (i) at least two misprints (ii) at least three misprints ?

Solution :

Since the probability p of occurrence of a misprint is very small ($p \rightarrow 0$) and the total number of words on a page is sufficiently large ($n \rightarrow \infty$) and we are given that $m = \text{average number of misprints per page} = \frac{100}{100} = 1$. The appropriate prob.

model to describe the given situation is the Poisson probability distribution with parameter $\lambda = 1$.

$$\begin{aligned}\text{Hence, } p(x) &= P(X = x) \\ &= P(\text{a page selected at random contains } x \text{ misprints}) \\ &= \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots \\ &= \frac{e^{-1} (1)^x}{x!} = \frac{1}{e} \cdot \frac{1}{x!}\end{aligned}$$

$$\begin{aligned}\text{(i) Req'd. prob. } P(X \geq 2) &= 1 - P(X \leq 1) \\ &= 1 - [p(0) + p(1)] \\ &= 1 - \left(\frac{1}{e} \cdot \frac{1}{0!} + \frac{1}{e} \cdot \frac{1}{1!}\right) \\ &= 1 - \frac{2}{e} = 1 - \frac{2}{2.71828} \\ &= 1 - 2 \times 0.3679 = 0.2642 \\ \text{(ii) Req'd. prob., } P(X \geq 3) &= 1 - P(X \leq 2) \\ &= 1 - [p(0) + p(1) + p(2)] \\ &= 1 - \frac{1}{e} \left(1 + 1 + \frac{1}{2}\right) \\ &= 1 - 0.3679 \times 2.5 \\ &= 0.08025\end{aligned}$$

Example 18.

A manufacturer finds that the average demand per day for the mechanics to repair his new product is 1.5, over a period of one year the demand per day is distributed as Poisson Variate. He employs two mechanics. On how many days in one year (a) both the mechanics would be free (b) some demand would be refused.

Solution :

Let a random variable X be the number of demands for a mechanic on any day. We are given that X follows Poisson distribution with parameter $\lambda = 1.5$.

Hence, the probability of r demands for a mechanic on any day is :

$$p(x) = P(X = x) = \frac{e^{-1.5} (1.5)^x}{x!}, \quad x = 0, 1, 2, \dots$$

- (a) Both the mechanics would be free, if there is no demand for any mechanic. Hence the required proportion of days on which no mechanic is used is given by

$$\begin{aligned}P(X = 0) &= e^{-1.5} \\ &= \text{antilog} [-1.5 \log_{10} e] \\ &= \text{antilog} [-1.5 \times \log_{10} 2.718] \\ &= \text{antilog} (-1.5 \times 0.4343) = \text{antilog} (-0.65145) \\ &= \text{antilog} (-1.34855) = 0.2231\end{aligned}$$

Hence, the expected number of days on which both mechanics would be free

$$= 365 \times P(X = 0) = 365 \times 0.2231 = 81.4315 = 81 \text{ days (approx)}$$

- (b) Since the manufacturer has employed only two mechanics, some demand will be refused if the number of demands per day is greater than 2. Hence the required proportion is given by

$$\begin{aligned}P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [p(0) + p(1) + p(2)] \\ &= 1 - e^{-1.5} \left(1 + 1.5 + \frac{(1.5)^2}{2!}\right) \\ &= 1 - 0.2231 (1 + 1.5 + 1.125) \\ &= 1 - 0.2231 \times 3.625 = 1 - 0.80874 = 0.19126\end{aligned}$$

Hence in one year, the expected number of days on which some demand will be refused $= 365 \times P(X > 2) = 365 \times 0.19126 = 69.81 = 70 \text{ days (approx.)}$

Example 19.

In a certain factory turning out blades, there is a small chance $1/500$ for one blade to be defective. The blades are supplied, in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective in a consignment of 10,000 packets. You are given that $e^{-0.02} = 0.9802$

Solution :

Here $N = 10,000$, $n = 10$, $p = 1/500 = \text{prob. of defective blade}$,

$$\lambda = np = 10 \times \frac{1}{500} = \frac{1}{50} = 0.02$$

Let the random variable X be the number of defective blades in a packet of 10. Then by Poisson probability distribution, the probability of x defective blades in a packet is given by

$$P(X = x) = \frac{e^{-0.04} (0.02)^x}{x!} = \frac{0.9802 (0.02)^x}{x!}$$

Hence in a consignment of 10000 packets the number of packets (frequency) containing r defective blades is :

$$N \times P(X = x) = \frac{10000 \times 0.9802 \times (0.02)^x}{x!}$$

Putting $x = 0, 1, 2, 3$, successively, we get :

$$\begin{aligned} \text{No. of packets containing no defective blade} &= 10000 \times 0.9802 \\ &= 9802 \end{aligned}$$

No. of packets containing 1 defective blade

$$\begin{aligned} &= \frac{10000 \times 0.9802 \times 0.02}{1!} \\ &= 9802 \times 0.02 = 196.04 = 196 \end{aligned}$$

No. of packets containing 2 defective blades

$$\begin{aligned} &= \frac{10000 \times 0.9802 \times (0.02)^2}{2!} \\ &= \frac{196.04 \times 0.02}{2} = 1.9604 = 2 \end{aligned}$$

No. of packets containing 3 defective blades

$$\begin{aligned} &= \frac{10000 \times 0.9802 \times (0.02)^3}{3!} \\ &= \frac{1.9609 \times 0.02}{3} = 0.0131 = 0 \text{ (nearly)} \end{aligned}$$

Note that the no. of packets can not be in fraction.

Example 20.

A systematic sample of 200 pages was taken from the manuscript typed by a typist and the observed frequency distribution of the typing mistakes per page was found to be as under :

No. of typing mistakes (X) :	0	1	2	3	4
No. of pages (f)	122	60	15	2	1

Fit the Poisson distribution to the above data.

Solution :

The mean of the given distribution is

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{122 \times 0 + 60 \times 1 + 15 \times 2 + 2 \times 3 + 1 \times 4}{200} = 0.5$$

Thus the mean of the Poisson distribution is $\lambda = \bar{X} = 0.5$.

By Poisson probability distribution, the theoretical frequencies are given by

$$f(x) = N.p(x) = 200 \times \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \dots$$

Put $x = 0, 1, 2, 3, 4$ successively in $f(x)$ we get the expected frequencies as:

Computation of expected frequencies

x	Expected Frequencies N.P (x)			
0	N.p (0)	$= 200 \times e^{-0.5}$	$= 200 \times 0.6064$	$= 121.28 \approx 121$
1	N.p (1)	$= N.p (0) \times \lambda$	$= 121.3 \times 0.5$	$= 60.65 \approx 61$
2	N.p (2)	$= N.p (1) \times \frac{1}{2}$	$= \frac{60.65 \times 0.5}{2}$	$= 15.3125 \approx 15$
3	N.p (3)	$= N.p (2) \times \frac{1}{3}$	$= \frac{15.3125 \times 0.5}{3}$	$= 2.552 \approx 3$
4	N.p (4)	$= N.p (3) \cdot \frac{1}{4}$	$= \frac{2.552 \times 0.5}{4}$	$= 0.32 \approx 0$

$$\begin{aligned} e^{-0.5} &= \text{antilog}(-0.5 \times \log_{10} e) \\ &= \text{antilog}(-0.5 \times 0.4343) \\ &= \text{antilog}(-0.21715) \\ &= \text{antilog}(\bar{1}.7828) = 0.6064 \end{aligned}$$

Note that the value of $e^{-0.5}$ can be obtained directly from any scientific calculator.

EXERCISE 2 (B)

1. Define Poisson probability distribution. What are its characteristics ? Mention three business situations where it is applicable.
2. If 3% of electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs exactly 5 bulbs are defective.
3. On the average one in 400 items is defective. If items are packed in boxes of 100, what is the probability that any given box of items will contain (i) no defectives (ii) less than two defectives (iii) one or more defectives (iv) more than three defectives
4. A manufacturer of pins knows that on an average 5% of his production is defective. He sells pins in boxes of 100, and guarantees that not more than 4 pins will be defective. What is the probability that (i) a box will meet the guaranteed quality (ii) will fail to meet the guarantee ?
5. The experience shows that 4 industrial accidents occur in a plant on an average per month. Calculate the probability of less than 3 accidents in a certain month. Use Poisson distribution (Given $e^{-4} = 0.01832$)
6. During a period, persons arrive at a railway booking counter at the rate of 30 per hour. What is the probability that two or fewer will arrive in a period of 5 minutes ? Use appropriate probability distribution.
7. In a tower 10 accidents took place in a span of 50 days. Assuming that the number of accidents per day follows the Poisson distribution, find the probability that there will be three or more accidents in a day.
8. If the chance of being killed by flood during a year is $1/3000$. Use Poisson distribution to calculate the probability that out of 3000 persons living in a village at least one would die in flood in a year.
9. An electric utility company has found that the weekly number of occurrence of lightning striking the transformers is a

Poisson distribution with mean 0.4. What is the probability that (i) no transformer will be struck in a week (ii) one transformer will be struck ?

10. The number of accidents in a year is attributed to taxi drivers in a city follows Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately the number of drivers with (i) no accidents in a year (ii) more than 3 accidents in a year. Given $e^{-1} = 0.3679$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$.
11. A firm produces articles of which 0.1 percent are defective and packs them in cases each containing 500 articles. A wholesaler purchases 100 such cases. In how many of them he expects to have no defective article and in how many of them 2 or more defectives.
12. The following table gives the number of days in 50-day period during which automobile accidents occurred in a certain part of a city. Fit a Poisson distribution to the data.

No. of accidents :	0	1	2	3	4
No. of days :	19	18	8	4	1

Also compute the mean and variance of the fitted distribution
13. The following table shows that distribution of the number of faulty units produced in a single shift in a factory. The data is for 100 shifts.

No. of faults :	0	1	2	3	4	5	6	and more
No. of shifts :	4	14	23	23	18	9	9	

Fit a Poisson distribution to the data.
14. Printing errors in the work produced by a particular firm occur randomly at an average rate of 0.6 per page. What is the probability that a seven-page pamphlet prepared by the firm contains more than three errors ?
15. Of the components produced by a certain firm 1.6 per cent are faulty. What is the probability that a box of 50 components contains two or more defectives ? by using (a) Binomial distribution (b) Poisson distribution
16. In a certain large factory the mean number of stoppages per week is 1.5. What is the probability that

- (i) in a given week there will be no stoppages:
- (ii) in a given week there will be three or more stoppages:
- (iii) in a given two-week period there will be at most one stoppage ?

2.4 NORMAL DISTRIBUTION

The normal distribution is the most important continuous probability distribution. It occupies an important place in management science for two reasons. First, it has properties that make it applicable to a number of managerial situations in which decision makers have to make inferences by drawing samples. Second, the normal distribution comes quite close to fitting the actual observed distribution of many natural processes including outputs from physical processes and human characteristics (height, weight, intelligence) as well as many other measures of interest to management in the social and natural sciences. Normal distribution is the *limiting case* of :

- (a) Binomial distribution when n is large and neither p nor q is very small.
- (b) Poisson distribution when its mean λ is large.

2.4.1 Definition of Normal Distribution

A continuous random variable X is said to be *normally distributed* if it has a probability density function (*p. d. f*) represented by the equation.

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty \quad (2-9)$$

where x = the value of the random variable.

μ = mean of the distribution of the random variable

σ = the standard deviation of the distribution

$\pi = 3.14159$

$e = 2.7183$

Note that the normal distribution is a continuous distribution with two *parameters* μ and σ and two *constants* π and e .

The graph of equation (9) is called the normal curve. It is a bell-shaped curve that extends indefinitely in both directions,

coming closer and closer to the horizontal axis without ever reaching it.

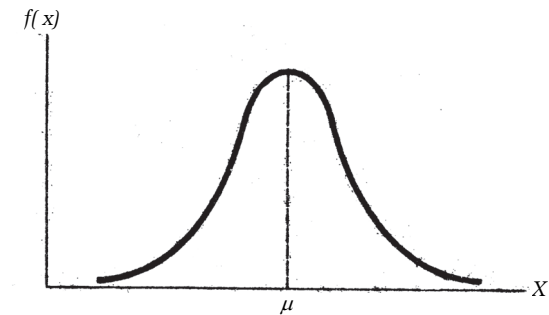


Fig. 2 – 1 : Normal Distribution

Remarks :

- (a) For any given pair of values μ and σ , there is one and only one normal distribution.
- (b) $X \sim N(\mu, \sigma^2)$ is used to denote the normal distribution with a mean μ and variance σ^2 .
- (c) The total area under the curve $p(x)$ is 1.
- (d) The probability that a normally distributed random variable with mean μ and variance σ^2 lies between two specified values a and b and is given by

$$P(a < X < b) = \text{Area under the curve } p(x) \text{ between } X = a \text{ and } X = b.$$

2.4.2 Standard Normal Distribution

Since a normal distribution is completely specified by its mean and standard deviation, there are different *bell-shaped* curves with different means and standard deviations. However it is possible to convert these bell-shaped curves or normal distributions into one standardised form called a *standard normal curve*. The areas under normal curve represent probabilities and can be obtained by using the tables prepared with reference to such standardised normal distributions in *Appendix 1*.

We have $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$, $-\infty < x < \infty$

Then the standard normal distribution or Z - distribution is expressed as

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; \quad -\infty < z < \infty \quad (2-10)$$

where $Z = \frac{X-\mu}{\sigma}$ = standard normal variable.

The standard normal distribution has zero mean and are standard deviation and hence it is denoted by $N(0, 1)$. The total area under the standardised normal distribution is 1. The probability that a standard normal variate lies between two specified values c and d is

$P(c < Z < d)$ = Area under the curve between two specified values
 $Z = c$ and $Z = d$.

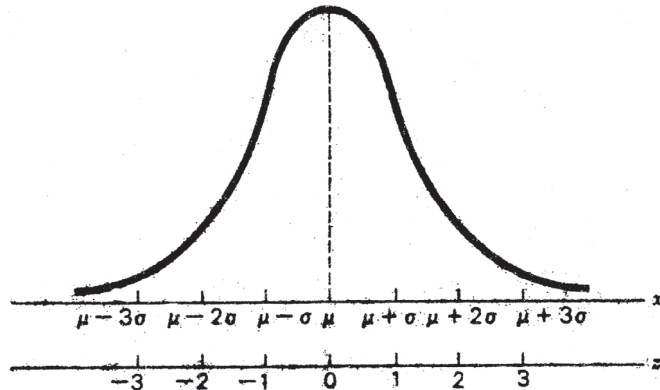


Fig 2 - 2 : Standard Normal Curve

2.4.3 Properties of Normal Distribution

The properties of a normal distribution are as follows :

1. It is perfectly symmetrical about the mean and is a bell shaped curve.
2. In a normal distribution, Mean = Median = Mode.
3. It is unimodal i.e. it has only one mode.

4. The ordinate at the mean of the normal curve divides the total area under the normal curve into two equal parts and each part is of area 0.5.
5. Some descriptive measures of a normal distribution are :
 - (i) Mean = μ , For standard form : $\mu = 0$
 - (ii) S.D. = σ , For standard form : $\sigma = 1$
 - (iii) It is a *symmetrical* or *mesokurtic* curve.
6. Quartiles Q_1 and Q_3 are equidistant from the median and are given by

$$Q_1 = m - 0.6745 \sigma$$

$$Q_3 = m + 0.6745 \sigma, \text{ where } m = \text{median}$$
7. Quartile deviation, $Q.D. = \frac{Q_3 - Q_1}{2} = 0.6745 \sigma = \frac{5}{6} M.D.$
 where $M.D.$ = mean deviation about the mean.
 Hence mean deviation about the mean is $\frac{4}{5} \sigma$.
8. As the distance of the curve from the mean increases, the curve approaches closer and closer to the axis but never touches it.
9. The maximum probability occurs at $X = \mu$ and the maximum value is $\frac{1}{\sigma\sqrt{2\pi}}$.

For larger value of σ the curve tends to flatten out and for smaller value of σ the curve has sharp peak.

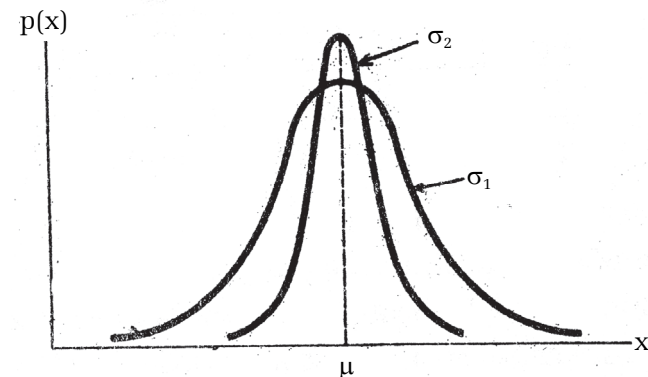


Fig 2 - 3 : Normal distributions $\sigma_1 > \sigma_2$

10. The sum of independent normal variables is also a normal variable.
11. Area property :
- The area under the normal curve between the ordinates at $X = \mu - \sigma$ and $X = \mu + \sigma$ is 0.6826. This implies that the range $\mu \pm \sigma$ occupies 68.26% of the observations.
 - The area under the normal curve between the ordinates at $X = \mu - 2\sigma$ and $X = \mu + 2\sigma$ is 0.9544. This implies that the range $\mu \pm 2\sigma$ occupies more than 95% of the observations.
 - The area under the normal curve between ordinates $X = \mu - 3\sigma$ and $X = \mu + 3\sigma$ is 0.9973. This implies the range $\mu \pm 3\sigma$ occupies (the entire area which is 1) almost all the observations
- It is important to note that :
- $\mu \pm 1.96 \sigma$ covers 95% of area (47.5% on each side).
- $\mu \pm 2.5758 \sigma$ covers 99% of area (49.5% on each side).
- $\mu \pm 0.6745 \sigma$ covers 50% of area (25% on each side).
12. The limiting case of the binomial distribution as $n \rightarrow \infty$ for fixed p is the normal distribution. If X is a binomial variable, $\frac{X - np}{\sqrt{npq}}$ is standard normal variable for large n . Note that for the binomial distribution, mean is np and standard deviation is $\sqrt{npq}$.

2.4.4 Importance of Normal Distribution

The following points disclose the importance of normal distributions.

- For large values of n , the computations of probabilities for discrete distributions become quite tedious and time consuming. In such cases, normal approximation of discrete distribution can be used with great ease and convenience.
- Many distributions which are not normal can be reduced to normal by simple transformation.

- It is essentially used in *large sampling theory* to find estimates of population *parameters* from sample *statistics* and *confidence limits* etc.
- Almost all the *exact sampling distributions* (e.g. Z - distributions, t - distributions, χ^2 - distributions, F - distribution) conform to normal distribution for large values of n (i.e. $n \rightarrow \infty$).

Example 21.

Ram and Co. manufactures chromed glass lamp manually. It requires 40 labour hours to complete one with a standard deviation of 10 hours.

- What is the probability that it will take between 35 and 42 labour hours to complete one of the lamps ?
 - What is the probability that it will take more than 48 hours ?
- (Given $Z_{0.2} = 0.0795$, $Z_{0.5} = 0.1915$, $Z_{0.8} = 0.2881$)

Solution :

Let the random variable X be the labour hours. Then X follows the normal distribution with mean $\mu = 40$ labour hours and s.d. $\sigma = 10$ hours.

$$\text{We have } Z = \frac{X - \mu}{\sigma}$$

$$\text{When } X = 35, Z = \frac{35 - 40}{10} = -0.5$$

$$\text{When } X = 42, Z = \frac{42 - 40}{10} = 0.2$$

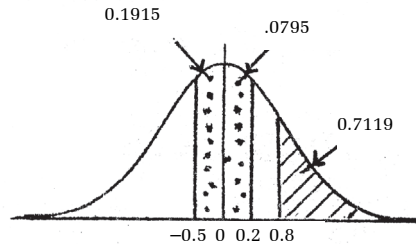
$$\text{When } X = 48, Z = \frac{48 - 40}{10} = 0.8$$

- Prob. that it will take between 35 and 42 labour hours to complete one of the lamps is given by

$$\begin{aligned} P(35 < X < 42) &= P(-0.5 < Z < 0.2) \\ &= P(-0.5 < Z < 0) + P(0 < Z < 0.2) \\ &= P(0 < Z < 0.5) + P(0 < Z < 0.2) \\ &= 0.1915 + 0.0795 = 0.2710 \end{aligned}$$

(ii) Prob. that it will take more than 48 hours is :

$$\begin{aligned} P(X > 48) &= P(Z > 0.8) = 0.5 - P(0 < Z < 0.8) \\ &= 0.5 - 0.2881 = 0.7119 \end{aligned}$$



Example 22.

The average daily sales of 500 branch offices was Rs. 150 thousand and the standard deviation Rs 15 thousand. Assuming the distribution to be normal indicate how many branches have sales between

- (i) Rs. 120 thousand and Rs. 145 thousand,
- (ii) Rs 140 thousand and Rs 165 thousand.

Solution :

Let the random variable X be the daily sales in rupees (in thousands) in a branch office. Then we are given that X is normally distributed with mean $\mu = 150$ and standard deviation $\sigma = 15$ and $N = 500$.

- (i) Prob. that the sales in a branch office lies between Rs. 120 thousand and Rs 145 thousand is given by : $P(120 \leq X \leq 145)$.

$$\begin{aligned} \text{The standard normal variable } Z \text{ is given by } Z &= \frac{X - \mu}{\sigma} \\ &= \frac{X - 150}{15} \end{aligned}$$

$$\text{When } X = 120, \quad Z = \frac{120 - 150}{15} = -2$$

$$\text{When } X = 145, \quad Z = \frac{145 - 150}{15} = -\frac{1}{3} = -0.33$$

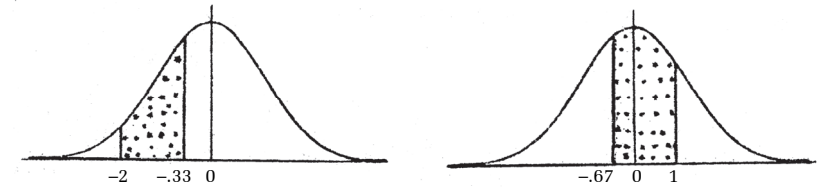
$$\begin{aligned} \therefore P(120 \leq X \leq 150) &= P(-2 \leq Z \leq -0.33) \\ &= P(0.33 \leq Z \leq 2) \\ &= P(0 \leq Z \leq 2) - P(0 \leq Z \leq 0.33) \\ &= 0.4772 - 0.1293 = 0.3479 \end{aligned}$$

(Read prob. values from Appendix 1)

Hence the no. of branches having sales between Rs 120 thousand and Rs. 145 thousand = $500 \times 0.3479 = 173.95 \approx 174$

$$\begin{aligned} \text{(ii) } P(140 \leq X \leq 165) &= P\left(\frac{140 - 150}{15} \leq Z \leq \frac{165 - 150}{15}\right) \\ &= P(-0.67 \leq Z \leq 1) \\ &= P(-0.07 \leq Z \leq 0) + P(0 \leq Z \leq 1) \\ &= P(0 \leq Z \leq 0.67) + P(0 \leq Z \leq 1) \\ &= 0.2486 + 0.3413 = 0.5899 \end{aligned}$$

Hence the number of branches having sales between Rs. 140 thousand and Rs. 165 thousand = $500 \times 0.5899 = 294.95 \approx 295$



Example 23.

The mean weight of 500 students at a certain campus is 65.6 kg and the standard deviation is 10 kg. Assuming that the weights are normally distributed find how many students weigh

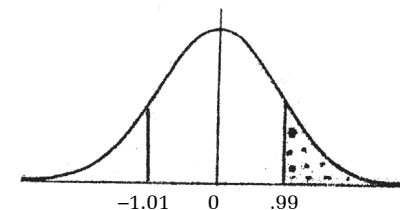
- (i) more than 75.5 kg (ii) between 55.5 and 75.5 kg

Solution :

Let X be the weight (in kg.) of students in the campus. Then X is normally distributed with mean $\mu = 65.6$ and S.D., $\sigma = 10$ and $N = 100$.

We have the standard normal variate, $Z = \frac{X - \mu}{\sigma}$

$$\text{When } X = 75.5, Z = \frac{75.5 - 65.6}{10} = 0.99$$



When $X = 55.5$, $Z = \frac{55.5 - 65.6}{10} = -1.01$

$$\begin{aligned} \text{(i) } P(X > 75.5) &= P(Z > 0.99) \\ &= 0.5 - P(0 < Z < 0.99) \\ &= 0.5 - 0.3389 = 0.1611 \end{aligned}$$

$\therefore$ the no. of students weighing more than 75.5 kg
 $= 500 \times 0.1611 = 80.55 = 81$ (nearly)

$$\begin{aligned} \text{(ii) } P(55.5 < X < 75.5) &= P(-1.01 < Z < 0.99) \\ &= P(-1.01 < Z < 0) + P(0 < Z < 0.99) \\ &= 0.3438 + 0.3389 = 0.6827 \end{aligned}$$

$\therefore$ the no. students weighing between 55.5 kg and 75.5 kg
 $= 500 \times 0.6827 = 341.35 \approx 341$ (nearly)

Example 24.

As result of tests of 2000 electric bulbs manufactured by a company it was found that life time of the bulbs was normally distributed with an average life of 2040 hours and a standard deviation of 60 hours. On the basis of this information, estimate the number of bulbs that is expected to burn for

(a) more than 2150 hours (b) less than 1960 hours.

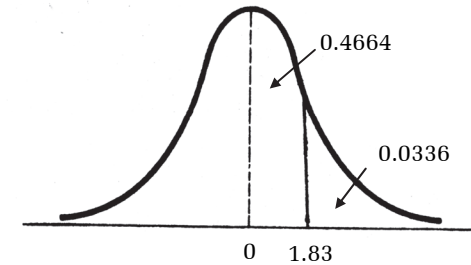
Z	Area	Z	Area	Z	Area
1.23	0.3907	1.33	0.4082	1.43	0.4236
1.63	0.4484	1.73	0.4582	1.83	0.4664

Solution :

Let X be the life of the bulbs in hours. Then X has a normal distributions with mean $\mu = 2040$ and S.D., $\sigma = 60$ and $N = 2000$.

(a) The prob. that the bulbs burn more than 2150 hours is given by

$$\begin{aligned} P(X > 2150) &= P\left(Z > \frac{2150 - 2040}{60}\right); \text{ where } Z = \frac{X - \mu}{\sigma} \\ &= P(Z > 1.83) = 0.5 - P(0 < Z < 1.83) \\ &= 0.5 - 0.4664 = 0.0336 \end{aligned}$$

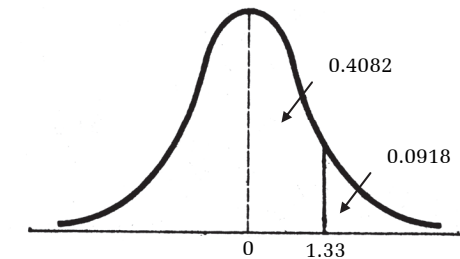


Hence the no. of bulbs that are expected to burn for more than 2150 hours $= 2000 P(X < 2150) = 2000 \times 0.0336 = 67.2 \approx 67$

(b) Prob. that the bulbs burn for less than 1960 hours is :

$$\begin{aligned} P(X < 1960) &= P\left(Z < \frac{1960 - 2040}{60}\right); \text{ where } Z = \frac{X - \mu}{\sigma} \\ &= P(Z < -1.33) = P(Z > 1.33) \\ &= 0.5 - P(0 < Z < 1.33) \\ &= 0.5 - 0.4082 = 0.0918 \end{aligned}$$

Hence the expected no. of bulbs to burn for less than 1960 hours $= 2000 P(X < 1960) = 2000 \times 0.0918 = 183.6 \approx 184$.



Example 25.

In an examination 15% of the candidates got first class (60% or above) while 40% failed (securing below 40%). Assuming the marks to be normally distributed, estimate the mean and the standard deviation.

Z at probability 0.1 = 0.25; Z at probability 0.4 = 1.28

Z at probability 0.35 = 1.04; Z at probability 0.15 = 0.385

Solution :

Let X be the marks in statistics.

Then $X \sim N(\mu, \sigma^2)$

By question, $P(X > 60) = 0.15$

$P(X < 40) = 0.40$

When $X = 60$, $Z = \frac{60 - \mu}{\sigma}$

When $X = 40$, $Z = \frac{40 - \mu}{\sigma}$

Hence from the given table,

$$\frac{60 - \mu}{\sigma} = 0.385 \quad (1)$$

$$\frac{40 - \mu}{\sigma} = -1.28 \quad (2)$$

$$\text{Divide (1) by (2)} \quad \frac{60 - \mu}{40 - \mu} = \frac{0.385}{-1.28} = -0.3$$

$$\text{or, } 60 - \mu = -12 + 0.3\mu \quad \text{or, } 1.3\mu = 72$$

$$\therefore \mu = \frac{72}{1.3} = 55.385 \approx 55$$

Substitute this value of μ in (1):

$$\frac{60 - 55.385}{\sigma} = 0.385 \quad \therefore \sigma = \frac{4.615}{0.385} = 11.99$$

Hence mean is 55 and s.d. is 12.

Example 26.

Given a normal distribution with $\mu = 50$ and $\sigma = 10$, find the value of X that has

- (i) 13% of the area to its left
- (ii) 0.4% of the area to its right.

Solution :

It is required to find the values

of X say x_1 and x_2 such that

$$P(X < x_1) = 0.13 \quad (1)$$

$$P(X > x_2) = 0.004 \quad (2)$$

where $X \sim N(\mu, \sigma^2)$; $\mu = 50$, $\sigma = 10$

These areas are shown in the figure. In the figure we observe that

$$\text{When } X = x_1, Z = \frac{x_1 - \mu}{\sigma} = -z_1$$

$$\text{When } X = x_2, Z = \frac{x_2 - \mu}{\sigma} = z_2$$

$$\text{Hence, we have, } \frac{x_1 - 50}{10} = -z_1 \quad (3)$$

$$\frac{x_2 - 50}{10} = z_2 \quad (4)$$

The relations (1) and (2) can be expressed as :

$$P(Z < -z_1) = 0.13 \quad (5)$$

$$P(Z > z_2) = 0.004 \quad (6)$$

From (5); $P(Z > z_1) = 0.13$ or, $0.5 - P(0 < Z < z_1) = 0.13$

$$\therefore P(0 < Z < z_1) = 0.5 - 0.13 = 0.37$$

$$\therefore z_1 = 1.13 \text{ (from Appendix 1)}$$

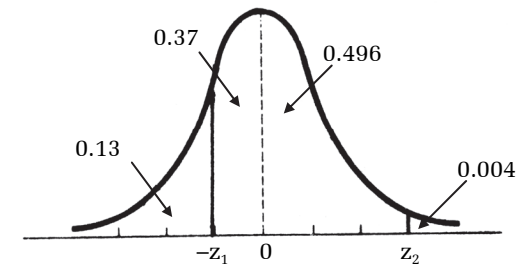
From (6); $0.5 - P(0 < Z < z_2) = 0.004$

$$\therefore P(0 < Z < z_2) = 0.5 - 0.004 = 0.496$$

$$\therefore z_2 = 2.65 \text{ (from Appendix 1)}$$

Substituting the values for z_1 and z_2 in (3) and (4);

$$x_1 = 50 - 10 \times 1.13 = 38.7 \text{ and } x_2 = 50 + 10 \times 2.65 = 76.5$$

**Example 27.**

Income of a group of 10000 persons were found to be normally distributed with mean Rs 520 and standard deviation Rs. 60. Find (i) the number of person having income between Rs 400 and Rs. 550 (ii) the lowest income of richest 1000 persons.

Area under the normal curve : 0.1915 0.3997 0.4015 0.4772

Value of Z : 0.5 1.28 1.29 2.00

Solution :

Let the random variable X be the income. Then X is normally distributed with mean $\mu = \text{Rs. } 520$ and s.d. (σ) = Rs. 60.

$$\text{When } X = 400, \quad Z = \frac{X - \mu}{\sigma} = \frac{400 - 520}{60} = \frac{-120}{60} = -2$$

$$\text{When } X = 550, \quad Z = \frac{X - \mu}{\sigma} = \frac{550 - 520}{60} = \frac{30}{60} = 0.5$$

$$\begin{aligned} \text{(i) } P(400 < X < 550) &= P(-2 < Z < 0.5) \\ &= P(-2 < Z < 0) + P(0 < Z < 0.5) \\ &= P(0 < Z < 2) + P(0 < Z < 0.5) \\ &= 0.4772 + 0.1915 = 0.6687 \end{aligned}$$

Hence the no. of person having income between Rs. 400 and Rs. 550 = $0.6687 \times 10000 = 6687$

$$\text{(ii) } P(X > x_1) = 1000/10000 = 0.1$$

where x_1 is the lowest income of richest 1000 persons.

$$\text{or, } P(Z > z_1) = 0.1, \text{ where } z_1 = x_1 - 520/60$$

$$\text{or, } 0.5 - P(0 < Z < z_1) = 0.1$$

$$\therefore P(0 < Z < z_1) = 0.4$$

$$\therefore z_1 = 1.28 \text{ (from the given table)}$$

$$\therefore x_1 - 520/60 = 1.28$$

$$\text{Hence } x_1 = 1.28 \times 60 + 520 = \text{Rs } 596.8$$

2.5 APPROXIMATIONS TO DISTRIBUTIONS**2.5.1 Normal Approximation to Binomial Distribution**

For large values of n , the binomial distribution can be approximated to the normal distribution. The approximation is valid when $np > 5$ or $n(1-p) > 5$ and this will be true for large n as p lies between 0 and 1. It may also be noted that for large values of n , calculation of binomial probabilities become cumbersome and so it is better to use the normal approximation.

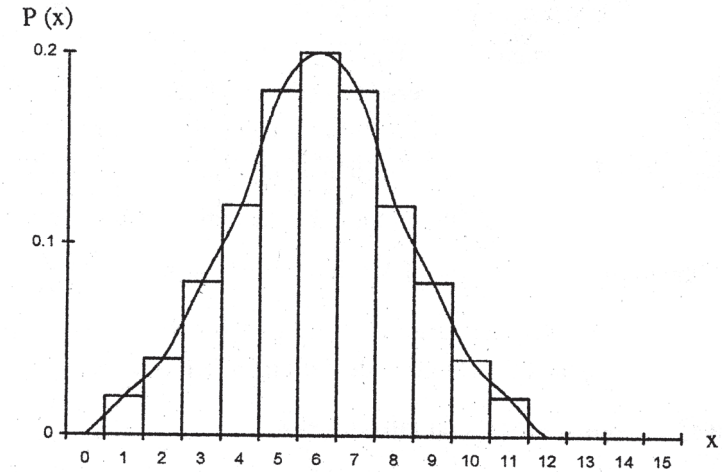


Fig 2 – 4 : Normal approximation to binomial

A typical normal approximation to binomial distribution looks as shown in Fig. 2.4 (The rectangles denote the binomial probabilities and the smooth curve is the corresponding normal curve.)

Once we decide to approximate the binomial to normal, then the standard variate for the approximated normal distribution will be

$$Z = \frac{X - np}{\sqrt{npq}} \quad \text{because } \bar{x} = np \text{ and } \sigma^2 = npq \text{ (for a binomial distribution)}$$

It should be noted while approximating discrete binomial distribution by a continuous normal distribution as mentioned above, a correction for continuity (continuity correction) is to be applied for improving the approximation. The binomial random variable X takes positive integer values such as 0, 1, 2 and considering the histogram representing the binomial distribution, the probability for any particular value, say $X = 3$, is nothing but the area of the rectangle containing the point $X = 3$. This rectangle spreads from $X = 2.5$ to $X = 3.5$. In a continuous curve, this area is to be approximated by the area under the normal curve between 2.5

and 3.5. That is, the probability for the binomial variable being 3 is to be approximated by the probability of the corresponding normal variable being between 2.5 and 3.5. Refer to Example 30 (b) to understand the concept more clearly

2.5.2 Normal Approximation to Poisson Distribution

If we have Poisson distribution where the mean number m of occurrences in an interval of the length being considered is large, then the Poisson probabilities can be approximated using a normal distribution with mean μ and standard deviation $\sqrt{\lambda}$. The approximation works adequately if $\lambda > 30$, and a continuity correction is needed since we are once more approximating a discrete distribution by a continuous distribution. (refer Example 28)

2.5.3 Poisson Approximation to Binomial Distribution

If we have a binomial experiment (i.e. n trials, two outcomes, and the same probability of success on each trial) where the number of trials is large compared with the number of success that could reasonably occur, the situation is like a Poisson one. The upper limit n on the number of successes is so large compared with what is likely to occur that it may as well be infinite.

Thus if we have a binomial situation where the probability of success p is less than 0.1 then the binomial probabilities can be approximated using the Poisson distribution with mean $\lambda = np$. This condition seems to be appropriate because it says that the mean number of successes per set of n trials is less than one-tenth of the possible number of successes, n . This is a useful approximation to make. Refer to solved Example 31.

Example 28.

The number of accidents occurring on average each year in a factory is 36. They occur completely at random. Use the normal approximation to the Poisson distribution to find the probability that in 1987 there were more than 40 accidents.

Solution :

$$\text{Here } \lambda = 36, \quad \therefore \text{S.D.} = \sqrt{\lambda} = \sqrt{36} = 6$$

Using normal approximation to the Poisson distribution, the probability that there were more than 40 accidents is given by

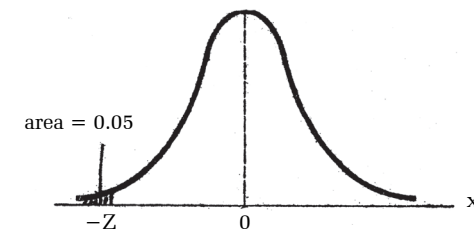
$$\begin{aligned} P(X > 40.5) &= P\left(Z > \frac{40.5 - 36}{6}\right) \\ &= P(Z > 0.75) \\ &= 0.5 - P(0 \leq Z \leq 0.75) \\ &= 0.5 - 0.2734 = 0.2266 \end{aligned}$$

Example 29.

A manufacturer of batteries wishes to give a guarantee for free replacement of the batteries whose life is less than a certain time period. If he desires to replace no more than 5% of the batteries, what should be the guarantee period, if the lives of the batteries are known to be normally distributed with mean of 1200 hours and a standard deviation of 100 hours ?

Solution :

The given information is indicated in the adjoining figure.



Here the value of x is to be determined.

$$\text{Given } P(Z < z_1) = 0.5 \text{ where } Z_1 = \frac{X - 1200}{100}$$

$$0.5 - P(-z_1 < Z < 0) = 0.05$$

$$P(-z_1 < Z < 0) = 0.5 - 0.05 = 0.45$$

$$\therefore -z_1 = 1.64 \text{ (value of } Z \text{ corresponding to area 0.45)}$$

$$\therefore \frac{X - 1200}{100} = -1.64 \quad \therefore X = 1200 - 164 = 1036 \text{ hours}$$

Thus a guarantee of 1036 hours may be given by the manufacturer.

Example 30.

- (a) An airline deliberately overbooks its local flights because it knows from experience that not all passengers who book for a given flight will actually arrive for that flight.

It is assumed that the probability of any booked passenger arriving for a given flight is 0.8; this is independent of the probability of any other passenger arriving.

The airline takes ten bookings for an eight-seater aircraft. Use the Binomial distribution to find the probability for a given flight:

- (i) that the aircraft takes off full;
 (ii) that the aircraft take off with at least two empty seats.
 (b) The independent probability of a passenger arriving for a booked flight on a flight is 0.8.

The airline books 225 passengers and there are 195 seats available on flight. Use a Normal approximation to find the probability that for a given flight more booked passengers arrive than there are seats available.

Solution :

- (a) Here $n = 10$, $p = 0.8$, $q = 1 - p = 0.2$
 (i) When the arrivals of booked passengers are 8 or 9 or 10, the aircraft takes off full.

Hence the required probability is :

$$\begin{aligned} & P(x = 8) + P(x = 9) + P(x = 10) \\ &= {}^{10}C_8 (0.8)^8 (0.2)^2 + {}^{10}C_9 (0.8)^9 (0.2) + (0.8)^{10} \\ &= (0.8)^8 [{}^{10}C_2 \times (0.2)^2 + {}^{10}C_1 (0.8) (0.2) + (0.8)^2] \\ &= 0.16777 (1.8 + 1.6 + 0.64) = 0.16777 \times 4.04 \\ &= 0.6778 \end{aligned}$$

- (ii) When the arrivals of booked passengers are 0 or 1 or 2 or 3 or 4 or 5 or 6, the aircraft takes off with atleast two empty seats. The required probability is given by :

$$\begin{aligned} & P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4) \\ & \quad + P(x = 5) + P(x = 6) \\ &= 1 - [P(x = 7) + P(x = 8) + P(x = 9) + P(x = 10)] \end{aligned}$$

$$= 1 - [{}^{10}C_7 (0.8)^7 (0.2)^3 + {}^{10}C_8 (0.8)^8 (0.2) + {}^{10}C_9 (0.8)^9 (0.2)^2 + (0.8)^{10}]$$

$$= 1 - [{}^{10}C_3 (0.8)^7 (0.2)^3 + 0.6778]$$

$$= 1 - [0.2013 + 0.6779] = 1 - 8791 = 0.1209$$

- (b) Here $n = 225$, $p = 0.8$, $q = 0.2$

$$\text{Mean} = np = 225 \times 0.8 = 180$$

$$\text{S.D.} = \sqrt{npq} = \sqrt{225 \times 0.8 \times 0.2} = \sqrt{36} = 6$$

We have to compute $P(X > 195)$

Using normal approximation to the binomial distribution the required probability is given by

$$\begin{aligned} & P\left(Z > \frac{180 - 195}{6}\right) \\ &= P(Z > -2.58) \\ &= P(Z > 2.58) \quad \text{By symmetry} \\ &= 0.5 - P(0 < Z < 2.58) \\ &= 0.5 - 0.4951 = 0.0049 \end{aligned}$$

Example 31.

Of the components produced by a certain form 1.6 per cent are faulty. What is the probability that a box of 50 components contains two or more defectives ? by using (a) Binomial distribution (b) Poisson distribution

Solution :

This is really a binomial problem. Each box is a set of $n = 50$ trials, and each trial has two outcomes in that a component can be either faulty or not faulty.

The probability of a success (i.e. a faulty component) is $p = 0.016$. The binomial calculation is

$$\begin{aligned} & P(2 \text{ or more defectives}) = [P(0 \text{ defectives}) + P(1 \text{ defective})] \\ &= 1 - \left[(1 - 0.016)^{50} + \frac{50!}{1! 49!} (0.016) (1 - 0.016)^{49} \right] \\ &= 1 - [(0.984)^{50} + 50 \times 0.016 \times (0.984)^{49}] \end{aligned}$$

$$= 1 - (0.446 + 0.363)$$

$$= 1 - 0.809 = 0.191$$

The Poisson approximation involves using the Poisson distribution with mean

$$\mu = np = 50 \times 0.016 = 0.8 \text{ Then}$$

$$P(2 \text{ or more defective}) = 1 - [P(0 \text{ defective}) + P(1 \text{ defective})]$$

$$= 1 - \left(\frac{0.8^0 e^{-0.8}}{0!} + \frac{0.8e^{-0.8}}{1!} \right)$$

$$= 1 - (0.449 + 0.359)$$

$$= 1 - 0.808 = 0.192$$

The two results are very similar, showing the approximation to be good for this case, where p is very small.

EXERCISE 2 (C)

- What is normal probability distributions. Explain in brief the properties of normal distribution.
- The mean weight of products is 68.22 grams with a variance of 10.8 grams. How many products in a batch of 1000 would you expect (i) to be over 72.0 grams (ii) between 70 and 72 grams. ($Z_{1.15} = 0.3749$, $Z_{.54} = 0.2054$)
- The weekly wages of 1000 workmen are normally distributed around a mean of Rs. 70 and with a standard deviation of Rs. 5. Estimate the number of workers whose weekly wages will be:
 - between Rs. 70 and Rs. 72
 - between Rs. 69 and 72
 - more than 75
 - less than 63
 - more than Rs. 80
 - less than 80
- The mean inside diameter of 500 washers produced by a machine is 5.02 mm. and the standard deviation is 0.05 mm. The purpose for which these washers are intended allows a maximum tolerance in the diameter of 4.96 to 5.08 mm, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine assuming the diameters are normally distributed.

- The marks obtained by the students in an examination are known to be normally distributed. If 10% of the students got less than 40 marks while 15% got over 80, what are the mean and standard deviation of marks ?
 - The marks obtained by 1000 students in an examination are known to be normally distributed. If 15% of the students got less than 30 marks and 10% of the students got over 90, find the mean and variance of the distribution.
- In a manufacturing organization the distribution of wages was perfectly normal and the number of workers employed in the organisation was 5,000. The mean wages of the workers were calculated as Rs. 800 p.m. and the standard deviation worked out to be Rs. 200. On the basis of the information, estimate :
 - The number of workers getting salary between Rs. 700 and Rs. 900.
 - Percentage of workers getting salary above Rs. 1000
 - Percentage of workers getting salary below Rs. 600.
- A sample of 100 dry battery cells tested to find the length of life produced the following results : Mean, $\bar{X} = 12$ hours, $\sigma = 3$ hours
Assuming the data to be normally distributed, what percentage of battery cells are expected to have life
 - more than 15 hours,
 - between 10 and 14 hours, and
 - less than 6 hours.
 - Kathmandu Municipality installed 2000 lights bulbs in the street of New Baneshwor. If these bulbs have average life of 1000 burning hours with s.d. of 200 hours, and, the life of bulbs are normally distributed, then how many bulbs are expected to fail in
 - less than 700 hours
 - more than 1200 hours
 - between 700 and 1300 ?
- Find the probability that the value of an item drawn at random from a normal distribution with mean 20 and standard deviation 10 will be between : (i) 10 and 15, (ii) – 5 and 10, and (iii) 15 and 25.

20. A project yields an average cash-flow of Rs. 500 lakhs with a standard deviation of Rs 60 lakhs. Calculate the following probabilities.
- Cash-flow will be more than Rs. 560 lakhs.
 - Cash-flow will be less than Rs. 420 lakhs.
 - Cash-flow will be between Rs. 460 lakhs and Rs. 540 lakhs.
 - Cash-flow will be more than Rs. 680 lakhs.
21. The customer accounts at a certain departmental store have an average balance of Rs. 480 and a standard deviation of Rs. 160. Assuming that the account balance are normally distributed.
- What proportion of the accounts is over Rs. 600 ?
 - What proportion of the accounts is between Rs. 400 and Rs. 600 ?
 - What proportion of the accounts is between Rs. 240 and Rs. 300 ?
22. The mean life of a certain type of electric light bulb is 1400 hours, with a standard deviation of 300 hours.
- If the manufacturer guarantees a life of 1000 hours, what percentage of bulbs can he expect to have returned ?
 - At what length of life should the guarantee be set in order for 95 per cent of bulbs to be found satisfactory ?
23. Applicants for a certain job are given an aptitude test. Past experience shows that the scores from the test are Normally distributed with a mean of 60 points and a standard deviation of 12 points. What per cent of candidates would be expected to pass the test, if a minimum score of 75 is required ?
24. A fair coin is tossed 100 times. Use the normal approximation to the binomial distribution to find the probability that the number of heads obtained is more than 45 but less than 60.
25. For a group of 1800 employees of a manufacturing company, IQ is approximately normally distributed with mean 110 and standard deviation 12. It is known from experience that for a particular job only persons with IQs of at least 95 are intelligent enough to do it, but that those with IQs greater than 120 soon become bored and unhappy with it. On the basis of

- IQ alone, how many of the 1800 employees would you expect to be suitable for the work ?
26. Steel bars produced at a machine have a mean length of 150 cm. with a standard deviation 0.2 cm. If bars having length between 149.6 cm. and 150.2 cm. are acceptable, determine, assuming normal distribution, the percentage of bars expected to be rejected. If the machine is re-adjusted such that the mean length changes to 149.9 cm. with no change in the standard deviation, what shall be the percentage of bars rejected ?
27. Assume that the marks in M.B.A examination are normally distributed with $\mu = 400$ and $\sigma = 100$ of 600 students taking this examination, it is desired to pass 500 of them, what should be the lowest marks permitted for passing ?
- | | | | | | |
|--------|--------|--------|--------|-------|--------|
| Z : | 0.95 | 0.96 | 0.97 | 0.44 | 0.45 |
| Area : | 0.3289 | 0.3315 | 0.3340 | 0.170 | 0.1736 |
28. In an examination, average marks secured by 400 students is 45 with a standard deviation of 10. Assuming the distribution to be normal find (i) the number of students securing marks between 50 and 60 (ii) the range of marks within which middle 50% of students would like